

GROKING AS A FIRST ORDER PHASE TRANSITION IN TWO LAYER NETWORKS

Noa Rubin*

Inbar Seroussi[†]Zohar Ringel[‡]

ABSTRACT

A key property of deep neural networks (DNNs) is their ability to learn new features during training. This intriguing aspect of deep learning stands out most clearly in recently reported Grokking phenomena. While mainly reflected as a sudden increase in test accuracy, Grokking is also believed to be a beyond lazy-learning/Gaussian Process (GP) phenomenon involving feature learning. Here we apply a recent development in the theory of feature learning, the adaptive kernel approach, to two teacher-student models with cubic-polynomial and modular addition teachers. We provide analytical predictions on feature learning and Grokking properties of these models and demonstrate a mapping between Grokking and the theory of phase transitions. We show that after Grokking, the state of the DNN is analogous to the mixed phase following a first-order phase transition. In this mixed phase, the DNN generates useful internal representations of the teacher that are sharply distinct from those before the transition.

1 INTRODUCTION

Feature learning is a process wherein useful representations are inferred from the data rather than being engineered. The success of deep learning is often attributed to this process. This is reflected, in part, by the performance gap between actual deep neural networks and their infinite-width Gaussian Process (GP) counterparts (Williams, 1996; Novak et al., 2018; Neal, 1996). It is also key to transfer learning applications (Weiss et al., 2016) and interpretability (Zeiler & Fergus, 2014; Chakraborty et al., 2017). Yet despite its importance, there is no consensus on how to measure or let alone classify feature learning effects.

Several recent results Li & Sompolinsky (2021); Ariosto et al. (2022) began shedding light on this matter. One line of work (adaptive kernel approaches) treats the covariance matrices of activation within each layer (kernels) as the key quantities undergoing feature learning. Feature learning would manifest as a deviation of these kernels from those of a random network and their adaptation to the task at hand. While providing a quantification of feature learning in quite generic settings, the equations governing these latent kernels are quite involved and may host a variety of learning phenomena. One such phenomenon (Seroussi et al., 2023), capable of providing a strong performance boost, is that of Gaussian Feature Learning (GFL) : A gradual process in which the covariance matrices of neuron pre-activations change during training so as to increase their fluctuations along label/target relevant directions. Remarkably, despite this smooth adaptation, the pre-activations' fluctuations, across width and training seeds, remain Gaussian. At the same time, the latent kernel itself develops notable spikes in the target direction, indicating feature learning.

Another phenomenon often associated with feature learning is Grokking. This abrupt phenomenon, first observed in large language models running on simple mathematical tasks, involves fast changes to the test accuracy following a longer period of constant and poor performance (Power et al., 2022). Though usually described as a time-dependent phenomenon, Grokking as a function of other parameters, it also occurs as a function of other parameters, such as sample size Power et al. (2022); Gromov (2023); Liu et al. (2022a). More broadly, DNNs behaviour as a function of time and dataset size

*Hebrew University, Racah Institute of Physics, Jerusalem, 9190401, Israel

[†]Department of Applied Mathematics, School of Mathematical Sciences, Tel Aviv University, Tel Aviv 69978, Israel

[‡]Hebrew University, Racah Institute of Physics, Jerusalem, 9190401, Israel

is often similar as reflected for instance the use of One Pass SGD You et al. (2014). Several authors provided quantitative explanations in the context of specific toy models wherein one can handcraft or reverse engineer the solution obtained by the network (Gromov, 2023; Nanda et al., 2023) or in suitably tailored perceptron models (Liu et al., 2022a; Žunkovič & Ilievski, 2022) where, however, representation learning is tricky to define. Given the aforementioned adaptive kernel approaches to deep learning, as well as the universality of Grokking across different DNNs and hyperparameters, it is natural to look for a more unifying picture of Grokking using a formalism that applies to generic deep networks.

In this work, we study Grokking as an equilibrium (or Bayesian) phenomenon driven by sample size, noise, or network width. Utilizing the aforementioned theoretical advancements, we show that Grokking in large-scale models can be classified and predicted through the mean field theory of phase transitions in physics (Landau & Lifshitz, 2013). Studying two different models, a teacher-student with cubic teacher and modular algebra arithmetic, we show the internal state of the DNN before Grokking is well described by GFL. In contrast, during Grokking, it is analogous to the mixed phase in the theory of first-order phase transitions, and the statistics of pre-activations are described by a mixture of Gaussian (GMFL). In this GMFL state, the latent kernels associated with the DNNs develop entirely new features that alter their sample complexity compared to standard infinite-width GP limits. After Grokking the weights are all specialized to the teacher. Besides providing a framework to classify feature learning effects, our approach provides analytically tractable and quantitative accurate predictions for the above two models.

Our main results are as follows:

- We establish a concrete mapping between the theory of first-order phase transitions, internal representations of DNNs, and Grokking for two non-linear DNN models each having two tunable layers.
- We identify three phases related to Grokking, one which is smoothly connected to the GP limit and two distinct phases involving different GP mixtures.
- For both our models, we simplify the task of learning high-dimensional representations to solving a non-linear equation involving either two (cubic teacher) or one (modular addition teacher) variables. Moreover, for the latter, we determine the location of the phase transition analytically.
- We flesh out a Grokking-based mechanism that can reduce the sample complexity compared to the GP limit.

Prior works. Phase transitions are ubiquitous in learning theory (e.g. Refs. Gardner & Derrida (1988); Seung et al. (1992); Györgyi (1990)), often in the context of replica-symmetry breaking. Connections between Grokking and phase transition were suggested by Nanda et al. (2023) but as far as analytic predictions go, prior work mainly focused on one trainable layer Žunkovič & Ilievski (2022); Arnaboldi et al. (2023) some suggesting those as an effective theory of representation learning Liu et al. (2022a). This can be further investigated by analyzing the loss landscape Liu et al. (2022b). Varma et al. (2023) suggest that the generalizing solution learned by the algorithm is more efficient but slower to learn than the memorizing one, using this interpretation they define regimes of semi-grokking, and ungrokking. The formalism of Refs. Arnaboldi et al. (2023); Saad & Solla (1995) can potentially be extended to online Grokking with two trainable layers, but would require reducing the large matrices involved. Phase transitions in representation learning have been studied in the context of bifurcation points in the information bottleneck approach (e.g. Tishby & Zaslavsky (2015)), nonetheless, the connection to deep learning remains qualitative. To the best of our knowledge, we provide a novel first-principals connection between grokking and phase transition *in representation learning*.

2 MODELS

2.1 NON-LINEAR TEACHER MODEL

Our first setting consists of a student Erf-network learning a single index non-linear teacher. The student is trained on a training set of size n , $\mathcal{D} = \{\mathbf{x}_\mu, y(\mathbf{x}_\mu)\}_{\mu=1}^n$ with MSE loss. In the following, bold symbol represents a vector. The input vector is $\mathbf{x}_\mu \in \mathbb{R}^d$ with iid Gaussian entries of variance

1. The target function y , is a scalar linear function of $\mathbf{x}$, with an orthogonal non-linear correction. Specifically, y is given by-

$$y(\mathbf{x}) = \underbrace{\mathbf{w}^* \cdot \mathbf{x}}_{H_1(\mathbf{w}^* \cdot \mathbf{x})} + \epsilon \underbrace{\left((\mathbf{w}^* \cdot \mathbf{x})^3 - 3|\mathbf{w}^*|^2 \mathbf{w}^* \cdot \mathbf{x} \right)}_{H_3(\mathbf{w}^* \cdot \mathbf{x})}. \quad (1)$$

where H_1, H_3 are the first two odd Hermite polynomials, and $\mathbf{w}^* \in \mathbb{R}^d$ are the teacher weights. For simplicity we take here the norm of the teacher weights to be 1, but this has no qualitative effect on the theory as long as we require $|\mathbf{w}^*| \sim \mathcal{O}(1)$. We consider a fully connected non-linear student network with one hidden layer of width N given by

$$f(\mathbf{x}) = \sum_{i=1}^N a_i \text{erf}(\mathbf{w}_i \cdot \mathbf{x}). \quad (2)$$

where $\mathbf{w}_i \in \mathbb{R}^d$ for $i \in [1, N]$ are the students weights. Evidence that this model Groks can be found in App. (D).

2.2 GROKING MODULAR ALGEBRA

Here we consider the setting of Ref. Gromov (2023), where the learning task is addition modulo P where P is prime. The network is trained on the following data set

$$\mathcal{D} = \{\mathbf{x}_{nm}, \mathbf{y}(\mathbf{x}_{nm}) \mid m, n \in \mathbb{Z}_P\} \quad (3)$$

where $\mathbf{x}_{nm} \in \mathbb{R}^{2P}$, is a vector such that it is zero in all its coordinates except in the coordinates n and $P + m$ where it is 1 (a ‘‘two-hot vector’’). The target function $\mathbf{y} \in \mathbb{R}^P$ is given by

$$y_p(\mathbf{x}_{nm}) = \delta_{p, (n+m) \bmod P} - 1/P, \quad (4)$$

where δ is the Kronecker delta and $\bmod P$ denotes the modulo operation which returns the remainder from the division by P . For the student model, we consider a two-layer deep neural network with a square activation function, given by

$$f_p(\mathbf{x}_{nm}) = \sum_{i=1}^N a_{pi} (\mathbf{w}_i \cdot \mathbf{x}_{mn})^2 \quad (5)$$

where $\mathbf{w}_c \in \mathbb{R}^{2P}$ for $c \in [1, N]$ are the students weights. For brevity, we denote $y_p(\mathbf{x}_{mn}) = y_{nm}^p$, and $f_p(\mathbf{x}_{nm}) = f_{mn}^p$.

2.3 TRAINING THE MODELS

In both cases, we consider networks that are trained with MSE loss to equilibrium using Langevin dynamics via algorithms such as Durmus & Moulines (2017); Neal et al. (2011). The continuum-time dynamics of the parameters are thus

$$\dot{\boldsymbol{\theta}}(t) = -\nabla_{\boldsymbol{\theta}} \left(\frac{\gamma}{2} \|\boldsymbol{\theta}(t)\|^2 + L(\boldsymbol{\theta}(t), \mathcal{D}) \right) + 2\sigma \boldsymbol{\xi}(t) \quad (6)$$

where $\boldsymbol{\theta}(t)$ is the vector of all network parameters in time t , γ is the strength of the weight decay, L is the loss function, the noise $\boldsymbol{\xi}$ is given by $\langle \xi_i(t) \xi_j(t') \rangle = \delta_{ij} \delta(t - t')$ and σ is the magnitude of the noise. We set the weight decay of the output layer so that with no data a_i^2, a_{pc}^2 both average to σ_a^2/N under the equilibrium ensemble of fully trained networks. The input layer weights are required to have a covariance of σ_w^2/d in the teacher-student model with $\sigma_w^2 = \mathcal{O}(1)$ and a covariance of 1 in the modular algebra model. Note that the covariance of the hidden layer is given by σ^2/γ . The posterior induced by the above training protocol coincides with that of Bayesian inference with a Gaussian prior on the weights defined by the above covariance and measurement noise σ^2 Naveh et al. (2021).

2.4 DERIVATION OVERVIEW

2.4.1 BRIEF INTRODUCTION TO MEAN FIELD THEORY PHASE TRANSITIONS

Phase transitions Landau & Lifshitz (2013); Tong (2011), such as the water-vapour transition, are ubiquitous in physics. They are marked by a singular behavior of some average observables as a function of a control parameter (say average density as a function of volume). As the laws of physics are typically smooth, phase transitions are inherently large-scale, or thermodynamic, phenomena loosely analogous to how the sum of many continuous functions may lead to a non-continuous one.

In a typical setting, the probability $p(x)$, of finding the system at a state x can be approximately marginalized to track a single random variable called an order parameter (Φ). As the latter is typically a sum of many variables (i.e. macroscopic), it tends to concentrate yielding a probability of the type $\log(p(\Phi)) \propto -d\tilde{S}(\Phi)$ where d is a macroscopic scale (e.g. number of particles) and $\tilde{S}(\Phi)$ is some well behaved function that does not scale with d . Given this structure, the statistics of Φ can be analyzed using saddle point methods, specifically by Taylor expanding to second order around minima of $\tilde{S}$.

Phase transitions occur when two or more *global* minima of $\tilde{S}(\Phi)$ appear. First-order phase transitions occur when these minima are distinct before the phase transition and only cross in $\tilde{S}$ value at the transition. Notably, the effect of such crossing is drastic since, at large d , the observed behaviour (e.g. the average Φ) would undergo a discontinuous change. Depending on the setup, such sharp change may be inconsistent as it would immediately change the constraints felt by the system. For instance, in the water-vapour transition, as one lowers volume the pressure on the vapour mounts. At some point, this makes a high-density minima of Φ as favourable as the low-density minima, signifying the appearance of water droplets. However, turning all vapor to droplets would create a drop in pressure making it unfavourable to form droplets. Instead, what is then observed is a mixture phase where as a function of the control parameter, a fraction of droplets forms so as to maintain two exactly degenerate minima of $\tilde{S}$. Lowering the volume further, a point is reached where all the vapour has turned into droplets and one is in the liquid phase.

In our analysis below, Φ will be a property of the weights in each neuron of the input layer, say their overlap with some given teacher weights. A high input dimension will be analogous to the large-scale limit, and the density loosely corresponds to the discrepancy in predictions. The phase transitions are marked by a new minima of $\tilde{S}(\Phi)$ which captures some feature of the teacher network useful in reducing the discrepancy. What we refer to as droplets corresponds to some input neurons attaining Φ values corresponding to the teacher feature while others fluctuate around the teacher agnostic minima. However the spatial notion associated with droplets is not relevant in this case as in the mean field theory of phase transitions.

2.4.2 ADAPTIVE KERNEL APPROACH AND ITS EXTENSION TO GAUSSIAN MIXTURES

Our main focus here is the posterior distribution of weights in the input layer ($p(\mathbf{w}_i)$) and posterior averaged predictions of the network ($f(\mathbf{x})$). Such posteriors are generally intractable for deep and/or non-linear networks. Hence, we turn to the approximation of Ref. Seroussi et al. (2023) where a mean-field decoupling between the read-out layer and the input layer is performed. This is exact in the limit of $N \rightarrow \infty$ and vanishing σ_a^2 (i.e. mean-field scaling). Following this, the posterior decouples to a product of two probabilities, a Gaussian probability for the read-out layer outputs and a generally non-Gaussian probability for the input layer weights. These two probabilities are coupled via two non-fluctuating quantities. The average kernel induced by the input layer and the discrepancy in predictions. As shown in Seroussi et al. (2023), specifically for a two-layer FCN, the resulting probability further decouples into iid probabilities over each neuron $p(\mathbf{w}_i)$. Below, we thus omit the neuron index i . The action ($-\log(p(\mathbf{w}_i))$ up to constant normalization factors) for each neuron’s weights is then given by the following form

$$\mathcal{S}[\mathbf{w}] = \frac{|\mathbf{w}|^2}{2\sigma_w^2} - \frac{\sigma_a^2}{2N} \sum_{\mu, \nu} \bar{\mathbf{t}}(\mathbf{x}_\mu)^T \bar{\mathbf{t}}(\mathbf{x}_\nu) \underbrace{\phi(\mathbf{w} \cdot \mathbf{x}_\mu) \phi(\mathbf{w} \cdot \mathbf{x}_\nu)}_{:= \sigma_a^{-2} \tilde{Q}_{\mu\nu}}, \quad (7)$$

where ϕ is the activation function and $\bar{\mathbf{t}}$ is the discrepancy between the averaged network output and the target given by

$$\bar{\mathbf{t}}(\mathbf{x}_\mu) = (\mathbf{y}(\mathbf{x}_\mu) - \bar{\mathbf{f}}(\mathbf{x}_\mu))/\sigma^2. \quad (8)$$

Notably $\bar{\mathbf{t}}$ is not given but determined by solving the following two mean-field self-consistency equations

$$\bar{\mathbf{f}} = Q [Q + \sigma^2 I_n]^{-1} \mathbf{y} \quad (9)$$

where the kernel Q is defined via

$$Q_{\mu\nu} = \sigma_a^2 \langle \phi(\mathbf{w} \cdot \mathbf{x}_\mu) \phi(\mathbf{w} \cdot \mathbf{x}_\nu) \rangle_{\mathcal{S}[\mathbf{w}]} \quad (10)$$

and $\langle \dots \rangle_{\mathcal{S}[\mathbf{w}]}$ denotes averaging over $\mathbf{w}$ with the probability implied by $\mathcal{S}[\mathbf{w}]$. In this work we use the equivalent kernel (EK) approximation, allowing the sums appearing in eqs. 7,9 to be replaced by integrals. This approximation washes out the generalization phenomena associated with Grokking, while capturing underlying feature learning mechanisms. As demonstrated in App. D, the feature learning effects hold also for finite datasets, in which a generalization gap can be observed. Theoretical corrections due to finite datasets can be made as shown in Cohen et al. (2021); Seroussi et al. (2023); Naveh & Ringel (2021), here we focus on the EK limit for simplicity.

Even if $\bar{\mathbf{t}}$ is given, the remaining action is still non-linear. Ref. Seroussi et al. (2023) proceed by performing a variational Gaussian approximation on that action. Here we extend this approximation into a certain variational Gaussian *mixture* approximation. Specifically, we show that as one scales up d, N, n in an appropriate manner, and following some decoupling arguments between, $\mathcal{S}[\mathbf{w}]$ has the form $d\hat{\mathcal{S}}[\Phi(\mathbf{w})]$ where $d \gg 1$ and $\hat{\mathcal{S}}[\Phi(\mathbf{w})]$ has $O(1)$ coefficients and $\Phi(\mathbf{w})$ is some linear combination of the weights. This allows us to treat the integration underlying $\langle \dots \rangle_{\mathcal{S}[\mathbf{w}]}$ within a saddle-point approximation. Notably when more than one Global saddle appears the saddle-point treatment corresponds to $p(\mathbf{w})$ having a Gaussian mixture form.

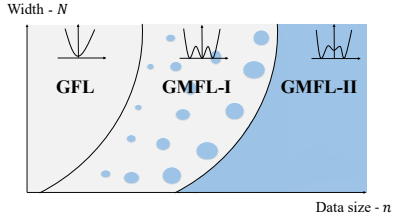


Figure 1: Schematic phase diagram of learning phases. The inset plots in the different phases correspond to the normalized negative log posterior ($\hat{\mathcal{S}}$). A transition between the different phases can be achieved by varying either n or N .

The three phases of learning described in the introduction correspond to the following behavior of the saddles (see also illustration in Fig. 1). At first, a single saddle centered around $\Phi(\mathbf{w}) = 0$ exists and weights fluctuate in a Gaussian manner around these minima, as in GFL. In the next phase, the distribution is comprised of a weighted average of this zero-saddle and other $|\Phi(\mathbf{w})| > 0$ saddles. This marks a new learning ability of the network and hence the beginning of Grokking. We name this mixture phase the first Gaussian Mixture Feature Learning phase (GMFL-I). This phase corresponds to the mixed (“droplets”) phase. In this phase, picking a random neuron, there is a finite chance it is in the GFL phase and hence fluctuates around the target agnostic minima at $\mathbf{w} = 0$. Similarly, there is a finite chance it fluctuates around one of the non-trivial saddles. At some later point,

after more data has been presented, only the saddles with $|\mathbf{w}| > 0$ dominate the average (GMFL-II). The phase phenomenology is shared by both models, however, the details of the new saddles that appear, as well as the decoupling scheme between the different components of $\mathbf{w}$ differ between both models. We next turn our attention to these details.

3 RESULTS

3.1 NON-LINEAR TEACHER STUDENT MODEL

Scaling setup and effective interaction. Consider the following two scaling variables (α, β) which we would soon take to infinity together and consider scaling up the microscopic parameters in the following manner

$$N \rightarrow \beta N \quad d \rightarrow \sqrt{\beta} d \quad \sigma_a^2 \rightarrow \sigma_a^2 / \sqrt{\beta} \quad , \quad \sigma^2 \rightarrow \frac{\alpha}{\beta} \sigma^2 \quad n \rightarrow \alpha n \quad (11)$$